

Mid Semester Examination

Course Name: Artificial Intelligence

Code: CS 366

Full Marks-50

Time: 2 hours

Answer ALL the questions

Make reasonable assumptions as and whenever necessary. You can answer the questions in any sequence. However, the answers to all the parts of any particular question should appear together.

Q1(a). Is sequence always important for classifying the document? What are the advantages of conditional independence assumption? What could be the demerits of this?

Q1(b). For a document classification problem, assume that the training set comprises the following five documents. The respective classes are also shown:

D1: {CPU, ALU, Classification}, Class=HW

D2: {Planning, ML, Logic}, Class= AI

D3: {Learning, Loss, Memory}, Class=AI

D4: {Paging, CPU, Classification}, Class= HW

D5: {Classification, ALU, ML }, Class= AI

Show how a multinomial naïve Bayes classifier will be estimated for the following document (*explain each step and decide which class to be assigned to the test document*)

D: {Classification, Learning, CPU, ML}

(2+2+2+ 10)=16

Q2 (a). Show the steps and compute the complexity of Bidirectional Search in case one side is BFS and the other side is DFS.

Q2(b). Consider a search space, where each state can be black, red, green, blue, and yellow. More than one state may have the same color. The target is to reach any black state. Some of rules on the successors of different states, based on their color:

- Any path between a red and a green state has cost ≥ 20 .
- Any path between a red and a blue state has cost ≥ 30 .
- Any path between a blue and a yellow state has cost ≥ 45 .
- Any path between a green and a yellow state has cost ≥ 50 .
- Any path between a red and a yellow state must go through a green or blue state.
- Any path connecting a red, green, or blue state to a black state must go through a yellow state.

Determine a heuristic, which is admissible in nature using the above information. H (Heuristics) should assign a value to a node based on the color of that node's state. That value should be the highest possible value that still guarantees that H is admissible.

(Show each step with proper intuition and explanation)

Q2(c). Show that A* Search always returns the optimal solution.

3+8+3 = 14

3(a). Consider the 8-puzzle problem (numbered from 1-8), where the goal state is 4-tiles away from the start state (i.e. 4 tiles are displaced from their original positions). Show how Hill climbing Search will follow the sequence to arrive at the destination from the start state.

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3(b). "Genetic Algorithm is a variation of Local Beam Search"- Explain in favor or against this claim with appropriate intuition and justification.

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3(c). Simulated annealing will find a global optimum with probability approaching 1 if you lower T slowly enough. Explain what simulated annealing will do in each of the following cases:

(i) What happens if you lower T very quickly?

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(ii) What happens if you never lower T?

Best of Luck

End Semester**Date:** April 29, 2024, **Time:** 0900 -1200 hrs (Morning Session)**Subject Code:** CS 366, **Subject Name:** Artificial Intelligence

Marks

- Q1(a). "First Order Logic is more expressive than Propositional Logic "- Explain this statement with an example. [3]
- Q1(b). Forward Chaining algorithm of inference could explore irrelevant paths as compared to the Backward chaining algorithm. Describe with proper intuition this statement. [3]
- Q1(c). Define definite clause and unit clause [2]
- Q2 Justify your answer for each statement.
- Q2(a). The resolution inference rule is complete but not sound. [3]
- Q2(b). Monotonicity is a desirable property for knowledge representation systems. [3]
- Q2(c). Propositionalization of First Order Logic Knowledge base may introduce perverse inference. [3]
- Q3. If a weight vector repeats during perceptron training, what can be concluded about its convergence? [3]
- Q4. Consider training the following NAND gate with the Perceptron algorithm.

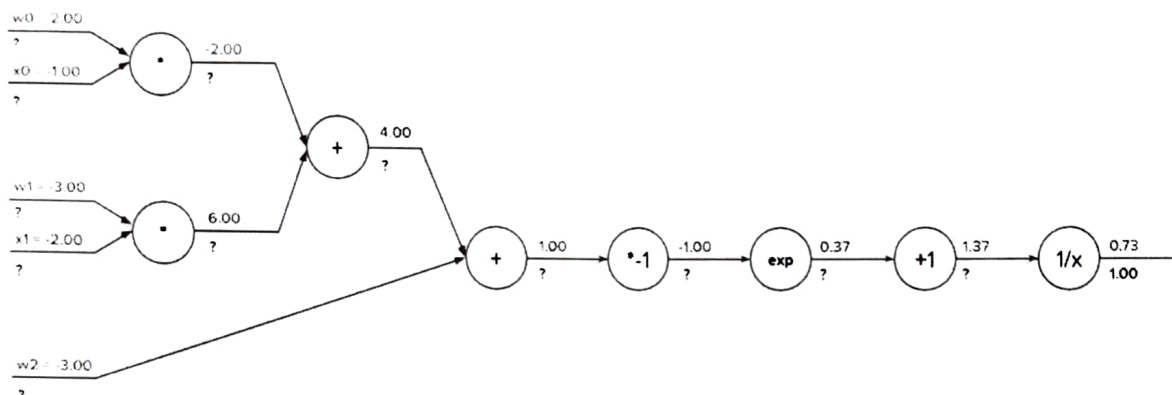
x1	x2	y
0	0	1
0	1	1
1	0	1
1	1	0

Consider the initial weights are $w_0 = -0.5$, $w_1 = 0.3$, and $w_2 = 0.2$ (where w_0 is the bias)
Assume learning rate = 1

- Q4(a). Show the Perceptron training steps [8]
- Q4(b). What would happen if $y=0$ for the first row? [2]
- Q5(a). Why is it necessary to include non-linearities in a neural network? Justify it Mathematically and using derivative. [4]
- Q5(b). A model was trained with the sigmoid activation function. The vanishing gradient problem was found to be present in it. Write an explanation of this using backpropagation. [2]

Q5(c). Describe when it is appropriate to use average pooling and maxpooling using example. [4]

Q6. The pre-activation function $z = w_0x_0 + w_1x_1 + w_2$ and the activation value $\alpha = \sigma(z) = 1 / (1 + e^{-z})$. The forward activations have been filled in the graph, on the top of the lines (Marked in Green colour), as well as the upstream gradient (Marked in Red colour) (gradient of the loss with respect to our neuron, $\partial L / \partial \alpha = 1$). Use this information to compute the rest of the gradients (labelled with question marks) throughout the graph. The computation graph is visualized below:

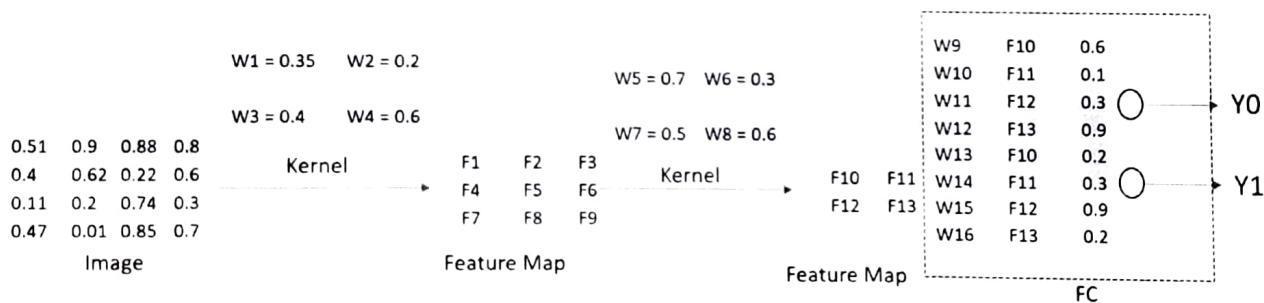


Finally, report the symbolic gradients with respect to the input parameters, x_0 , x_1 , w_0 , w_1 , and w_2 (Note: The mathematical equation and the values both need to be present in the solution)

1. $\partial \alpha / \partial x_0 =$	2. $\partial \alpha / \partial w_0 =$	3. $\partial \alpha / \partial x_1 =$	4. $\partial \alpha / \partial w_1 =$	5. $\partial \alpha / \partial w_2 =$
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Q7 A simple CNN model is used to class the image of a dog (Ground truth $\Rightarrow Y_0 = 1, Y_1 = 0$), The cross entropy loss was used in the model

$$-\sum_{c=1}^N y_c \log(p_c)$$



Q7(a). Compute the values of the feature map F10 to F13 in forward pass [4]

Q7(b). Compute the derivative of $W_5, W_6, W_7, W_8, F_{10}, F_{11}, F_{12}, F_{13}$ with respect to Loss. (The solution should contain the mathematical representation and the numerical value for the same) in backward pass. (Note Only for 1 Backward Pass) [16]

Note: The values should be rounded to two decimal places at each level.